Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*			Fuel is <u>burned</u> to heat <u>water</u> [to make steam] (1) [Kinetic energy] of <u>steam</u> turns or drives or rotates a <u>turbine</u> (1) [Turbine turns or drives] <u>generator</u> producing electricity (1) Don't accept moves or pushes.	3			3		
	(b)			Useful power = $\frac{3.2}{40} \times 100 = 8$ [MW] (1) 8 MW = 75 % (1) So total input = $\frac{8}{75} \times 100 = 10.7$ [MW] (1) so she is not correct - to award full marks the conclusion must be present **Alternative:** 40 % of 75 % = 30 % (1) 30 % = 3.2 [MW] (1) so total input = $\frac{3.2}{30}$ x 100 = 10.7 [MW] (1) so she is not correct - to award full marks the conclusion must be present **Alternative:** $\frac{75}{100}$ x 20 = 15 [MWh] (1) $\frac{40}{100}$ x 15 = 6 [MWh] (1) which is not 3.2 [MWh] (1) so she is not correct - to award full marks the conclusion must be present **Alternative:** $\frac{3.2}{20}$ x 100 = 16 [%] (1) 40 % of 75 % = 30 % (1)			3	3	3	

				16 is not equal to 30 % (1) so she is not correct - to award full marks the conclusion must be present						
	(c)			$I = \frac{P}{V}$ (1) manipulation $I = \frac{1200}{400}$ (1) substitution I = 3 000 [A] (1) answer Answer = 3×10^n where n is not 3 award 2 marks	 1	1 1		 3	 3	
				Question 5 total	**4**	**2**	**3**	**9**	**3**	**0**